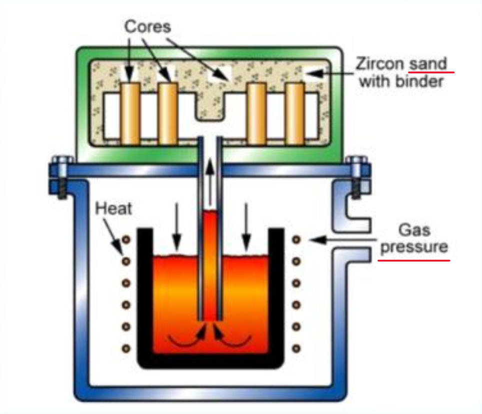
Week 7-Quiz

21 May 2025 18:05

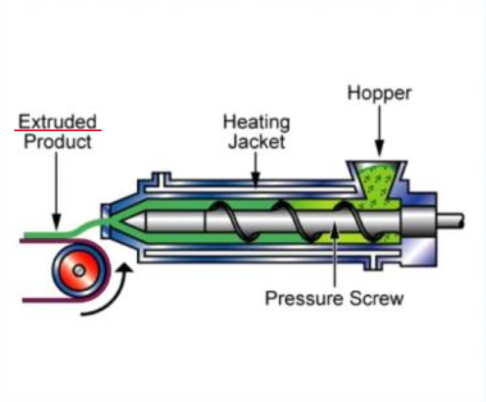
Which process is this?



Select one:

- a. Investment Casting
- b. Low Pressure Sand Casting
- c. Evaporative Pattern Sand Casting
- d. Gravity Die Casting

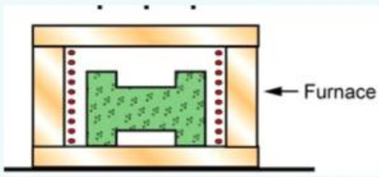
Which process is this?



Select one:

- a. Blow Moulding
- b. Extrusion
- c. Rotational Moulding
- d. Injection Moulding

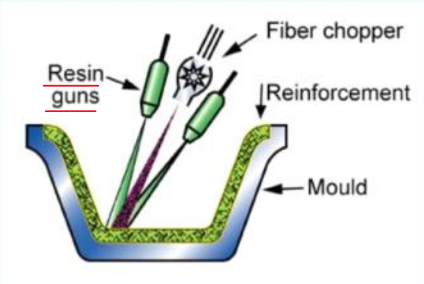
Which process is this?



Select one:

- a. Sintering
- b. Hot Isostatic Pressing
- c. Hot Pressing
- d. Tape Casting

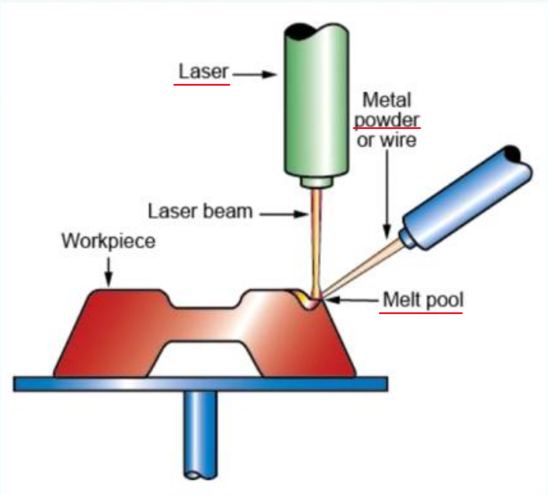
Which process is this?



Select one:

- a. Dough Moulding Compound Moulding
- b. Sheet Moulding Compound Moulding
- c. Filament Winding
- d. Spray Lay Up

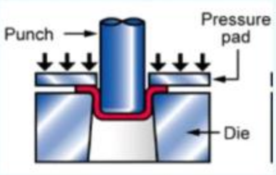
Which process is this?



Select one:

- a. Selective laser sintering
- b. Laser powder forming
- c. Laminated object manufacturing
- d. Stereolithography
- e. Electron beam melting
- f. Selective laser melting

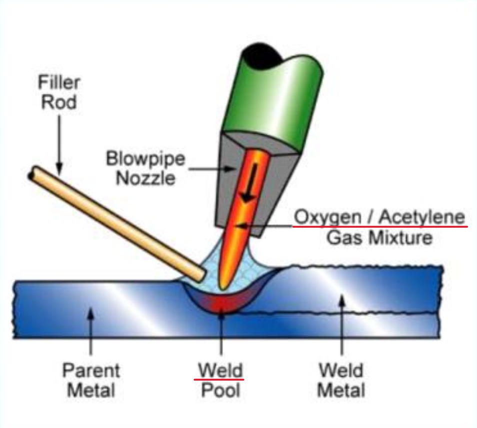
Which process is this?



Select one:

- a. Bending
- b. Deep Drawing
- c. Blanking
- d. Stretching

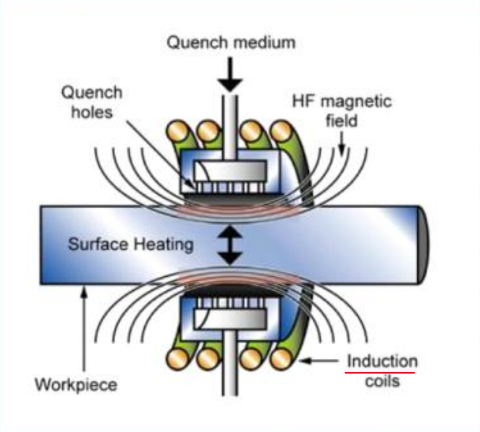
Which process is this?



Select one:

- a. Brazing
- b. Oxyacetylene welding
- c. Gas tungsten arc welding
- d. Gas metal arc welding
- e. Ultrasonic welding
- f. Power beam welding
- g. Friction Stir Welding

Which process is this?



Select one:

- a. Induction Hardening ✓
- b. Laser Hardening
- c. Flame Hardening
- d. Nitriding

Thrust deflectors and reversers on aircraft engines are controlled by hydraulic actuators. The forces required to reverse thrust can be large, so the actuators (of which each engine may have four) are heavy. One of the heavier parts is the piston rod, identified in the figure below. It carries axial loads only, and is designed as a hollow tube, to save weight. The cross section is chosen to carry the loads without compressive failure or buckling. Currently rods are made of high-strength steel. Could there be a better material?

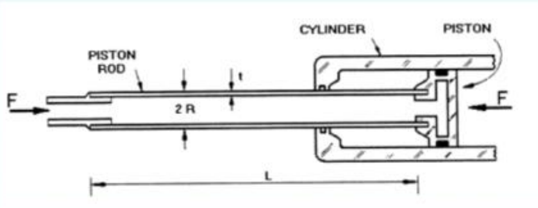


Figure 1. A Piston Rod

If we model the piston rod as a thin-walled tube of length *L*, radius *R* and wall thickness *t*. What is the mass of the rod?

$m = \rho V = \rho \times 2\pi R t L$

Select one:

- $m = 2\pi RtL\rho$ ✓
- $m = \pi R^2 L\rho$
- $m = \pi R^2 tL\rho$
- $m = \pi R^2 t\rho$

Specify the failure criterion for a yield failure to arise in terms of the rod dimensions and the strength of the material, σ_y used?

Select one:

$\sigma_y > \sigma = \frac{F}{A} = \frac{F}{2\pi Rt}$

- $\sigma_y > \frac{F}{\pi R^2 t}$
- $\sigma_y > \frac{F}{\pi Rt}$
- $\sigma_y > \frac{F}{2\pi Rt}$ ✓
- $\sigma_y > \frac{F}{\pi R^2}$

By eliminating the wall thickness from your two equations, calculate the mass of the piston rod in terms of the required materials properties and the specified length, L and maximum required force, F. This mass is the value that you would like to minimise.

$$\left.\begin{array}{l} m = 2\pi R t L \rho \\ 2\pi R t = \dfrac{F}{\sigma_y} \end{array}\right\} \quad m = L\rho \cdot \frac{F}{\sigma_y}$$

Select one:

- $mass = \frac{L}{F}\frac{\sigma_y}{\rho}$
- ✓ $mass = LF\frac{\rho}{\sigma_y}$
- $mass = LF\frac{\sigma_y}{\rho}$
- $mass = \frac{L}{F}\frac{\rho}{\sigma_y}$

M_1 is derived from the previous mass equation. We wanted to minimise the mass, but for materials index value it is common to invert the properties to create a materials index that is maximised. So examine the materials properties in the mass equation and then decide which of the following is the appropriate combination of materials properties that you would like to maximise as M_1?

Minimise : m $\Rightarrow$ ∴ Minimise : $\frac{\rho}{\sigma_y}$ or maximise : $\frac{\sigma_y}{\rho}$

Select one:

- $M_1 = \sigma_y^3/\rho$
- $M_1 = \sigma_y^2/\rho$
- ✓ $M_1 = \sigma_y/\rho$
- $M_1 = E/\rho$

Buckling occurs when an axially loaded strut is compressed by force, F

$F = \frac{\pi^2 EI}{L^2}$

where I is given for a hollow cylinder as

$I = \pi R^3 t$

Combine these equations and rearrange to make the wall thickness, t the subject and substitute this back into your original mass equation for the hollow cylinder. Write out the resulting equation for the mass of the cylinder.

$$F = \frac{\pi^2 E(\pi R^3 t)}{L^2} \Rightarrow t = \frac{FL^2}{\pi^3 E R^3}$$

$$m = 2\pi R L t \rho$$

$$m = \frac{2FL^3}{\pi^2 R^2} \cdot \frac{\rho}{E}$$

Select one:

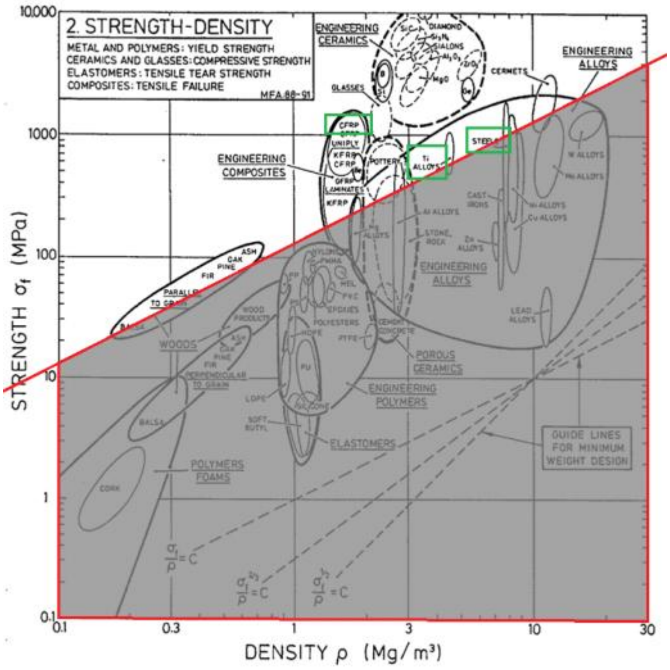
- ✓ $m = \frac{2FL^3}{\pi^2 R^2}\frac{\rho}{E}$
- $m = \frac{2FL^3}{\pi^2 R^2}\frac{E}{\rho}$
- $m = \frac{2FL^2}{\pi R^2}\frac{\rho}{E}$
- $m = \frac{2FL^2}{\pi R}\frac{E}{\rho}$

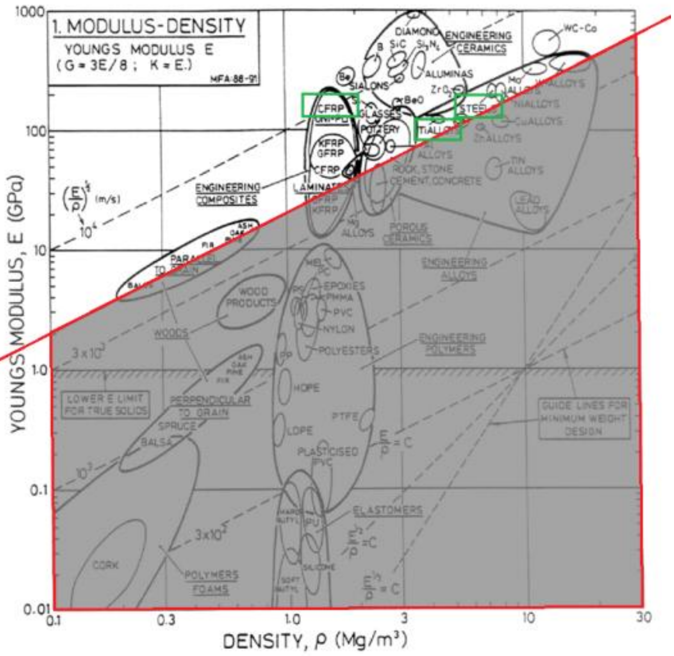
By looking back at the equation for the mass of the rod. Derive an alternative material performance index M_2 that you will want to maximise in order to identify the materials for a minimum mass strut that will not buckle under the maximum load.

Minimise : m $\Rightarrow$ Minimise : $\frac{\rho}{E}$ or Maximise : $\frac{E}{\rho}$

Select one:

- $M_2 = \frac{\sigma_y}{E}$
- $M_2 = \frac{\sigma_y}{\rho}$
- ✓ $M_2 = \frac{E}{\rho}$
- $M_2 = \frac{\rho}{\sigma_y}$





It is clear especially after recent climate summits and energy price volatility, that offshore wind farms are an increasingly important part of the energy supply chain in the UK. Offshore wind turbine towers have to perform one essential function and that is to support the turbine blades and generator. The primary function is that of a column in compression. The design must last at least 30 years, with little or no maintenance. The marine environment is highly corrosive and the mast has to be able to withstand the worst physical action that the environment will present over the lifetime of the turbine. One significant issue to overcome is the exposure to UV sunlight and the highly corrosive nature of the salt and below water there is the problem of fouling. Then the mast must withstand the extreme winds that will be exerienced over their lifetimes. The tower must also be tough enough to withstand the impact of waves and then finally it must be able to cope with both high and low operating temperatures. In practice the size (or Radius) of the mast is a free variable as the choice of material to make it.

If we set a series of screening criteria which of the following is the most sensible set of screening requirements for the initial narrowing down our search field.

Select one:

- a. Resisit UV, resist corrosion in marine environment, high toughness & must function from -40°C to 40°C
- b. Resisit UV, high toughness, must not fail in high winds & function from -40°C to 40°C
- c. Resisit UV, resist corrosion in marine environment, high toughness, must not buckle.

The final objective is to minimise the cost. The height of the tower is fixed. For simplicity presume that the tower is solid. Assuming that the tower is of length, L and radius R calculate the volume and hence the weight. Next you need to work out the cost of the tower which is the parameter to be optimised assuming that the cost per unit mass is known C_m. Which of the following expressions is the correct one for the cost of the material used in the tower.

Select one:

$m = \pi R^2 L \rho$

- $\pi R^2 L \frac{C_m}{\rho}$
- $\pi R^2 L \frac{\rho}{C_m}$
- $\pi R^2 L \rho C_m$
- $\frac{\pi R^2 L}{C_m \rho}$

$Cost = m C_m = \pi R^2 L \rho C_m$

The tower needs to be able to withstand a maximum bending moment M_c, without exceeding the yield strength, σ_y of the materials. You can assume that for a solid cylinder that:

$$M_c = \frac{\pi R^3 \sigma_y}{4}$$

Rearrange this equation and substitute the radius back into the equation for the cost of the mast. Which of the following equations is the correct one for the cost of the tower now expressed in terms of the fixed variables and the materials properties.

$$R = \left(\frac{4M_c}{\pi \sigma_y}\right)^{\frac{1}{3}}$$

$$Cost = \pi \left(\frac{4M_c}{\pi \sigma_y}\right)^{\frac{2}{3}} L \rho C_M$$

Select one:

- a. $C_t = \pi^{3/2}(\frac{\sigma_y}{4M_c})^{1/2} L\rho C_m$
- b. $C_t = \pi^{1/3}(\frac{4M_c}{\sigma_y})^{2/3} L\rho C_m$ ✓
- c. $C_t = \pi^{5/3}(\frac{\sigma_y}{4M_c})^{2/3} L\rho C_m$
- d. $C_t = \pi^{2/3}(\frac{4M_c}{\sigma_y})^{1/3} L\rho C_m$

Looking at the previous answer, determine the materials index that we can derive for the lowest cost tower of a specified height that that can take a particular maximum bending moment. This function should be minimised.

Minimise : Cost $\Rightarrow$ Minimise : $\frac{\rho C_m}{(\sigma_y)^{2/3}}$

Select one:

- $M_1 = \frac{\rho C_m}{\sigma_y^{2/3}}$ ✓
- $M_1 = \frac{C_m}{\rho \sigma_y^{3/2}}$
- $M_1 = \frac{\rho C_m}{\sigma_y^{3/2}}$
- $M_1 = \frac{C_m}{\rho \sigma_y^{2/3}}$

A 90g polymer component can be manufactured by either machining from an extruded bar or by injection moulding from extruded pellets. Ignoring the capital and labour costs whilst assuming the following information.

Polymer material cost - £8.00 /kg

The waste material created per part by machining is 160g.

The amount of waste material created per part by injection moulding is 10g.

The tooling required to machine these parts is £1,000.

The injection moulding tooling costs dedicated for moulding this part are - £25,000

What is the cost in £ per part of machining 1,000 of these parts?

Answer: £3.00

Cost per unit, C (without extra stuff) =	3
mass per unit, m (kg) =	0.09
Waste mass per unit (kg) =	0.16
Total mass per unit (kg) =	0.25
Scrap fraction, f =	0.64
Material costs, C_m (£/kg) =	8
Tooling cost, C_t (£) =	1000
Batch size, n =	1000

What is the cost (in £) per part of injection moulding 1000 of these parts?

Answer: £25.80

Cost per unit, C (without extra stuff) =	25.8
mass per unit, m (kg) =	0.09
Waste mass per unit (kg) =	0.01
Total mass per unit (kg) =	0.1
Scrap fraction, f =	0.1
Material costs, C_m (£/kg) =	8
Tooling cost, C_t (£) =	25000
Batch size, n =	1000

What is the cost per part (in £) of machining 100,000 of these parts?

Answer: 2.01

What is the cost per part (in £) of injection moulding 100,000 of these parts?

Answer: 1.05

What manufacturing number is the cross over point where the cost per part is equivalent for both methods of manufacture?

Answer:

Equation 1:

$$C = 2 + \frac{1000}{n}$$

=

Equation 2:

$$C = 0.8 + \frac{25000}{n}$$

$$\frac{24000}{n} = 1.2$$

$$n = \frac{24000}{1.2} \approx 20000$$